

EDA SUMMARY & CRITICAL FINDINGS
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Total rows: 32509 | Stations: 6
Date range: 2021-02-23 21:00:00 to 2022-06-21 17:00:00
PM2.5: Mean=22.6, Median=18.6 (Right-Skewed)

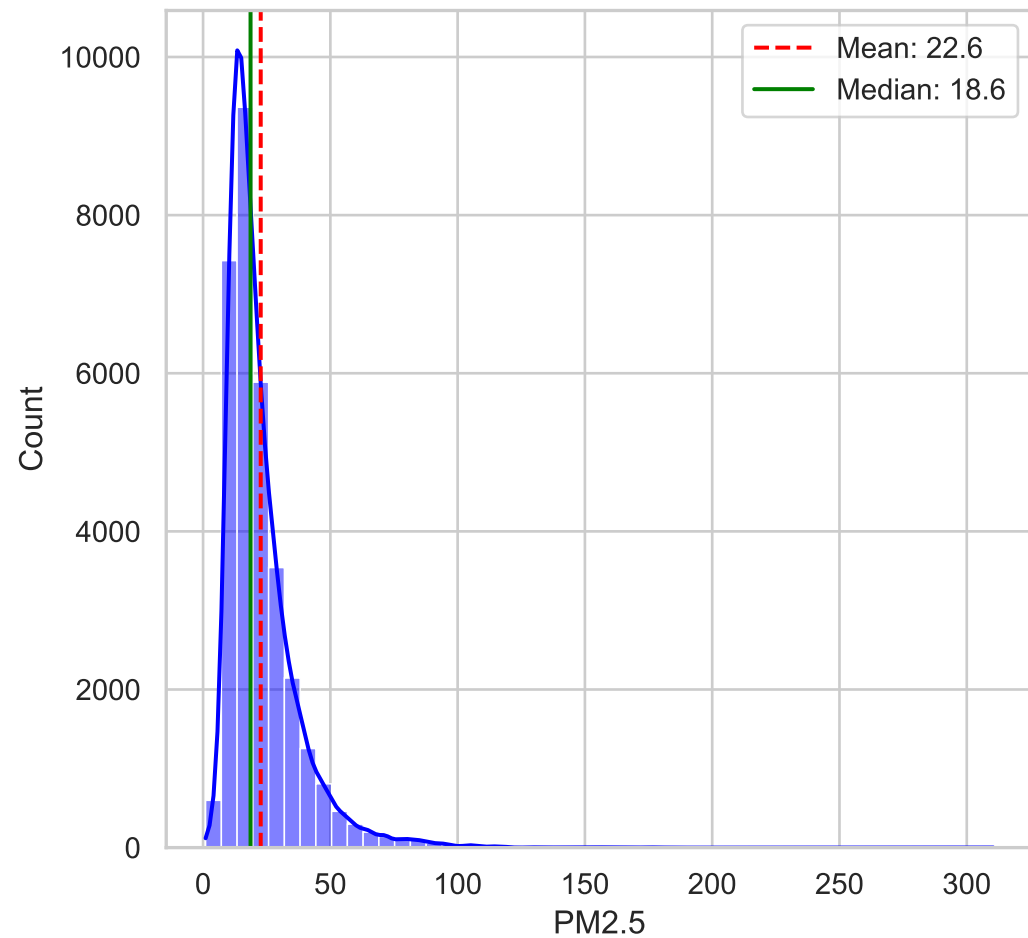
CRITICAL DATA QUALITY WARNINGS:

- ! OUTLIER CO: Max 10809 (Station 2, 10/03/2021) - ~27x mean.
- ! SO2 ANOMALY: Mean 229.3 - Check units. Action: Multiply by 2.62 (if ppb) or exclude if unverified.
- ! TSP ISSUE: 8031 zero values (24.7%) - ACTION: Use timeline analysis to verify sensor offline.
- ! STATION BIAS: Station 1 & 2 ended early. ACTION: Consider excluding or weighted sampling.

KEY INSIGHTS:

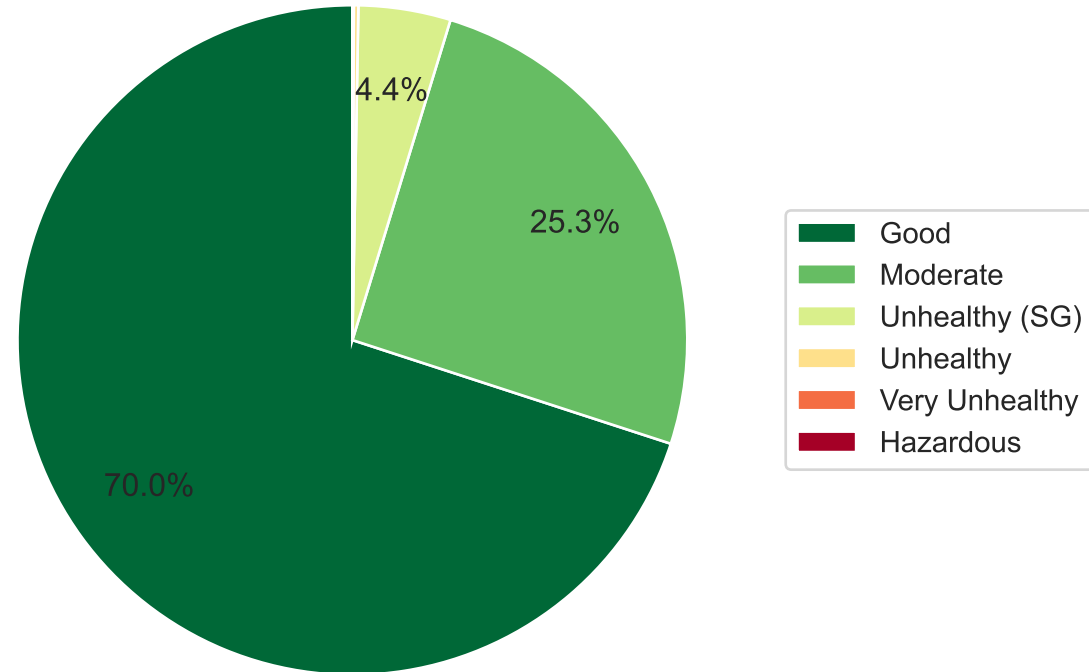
- * PM2.5 peaks at 5-8 AM (rush hour + atmospheric stability).
- * Dry season (Nov-Jan) pollution is ~2x higher than rainy season.
- * Strong CO/SO2 correlation (0.66) indicates shared combustion sources.
- * Station 4 is the most polluted (Mean 26.3), Station 5 is cleanest (16.8).

Distribution of PM2.5 Concentrations

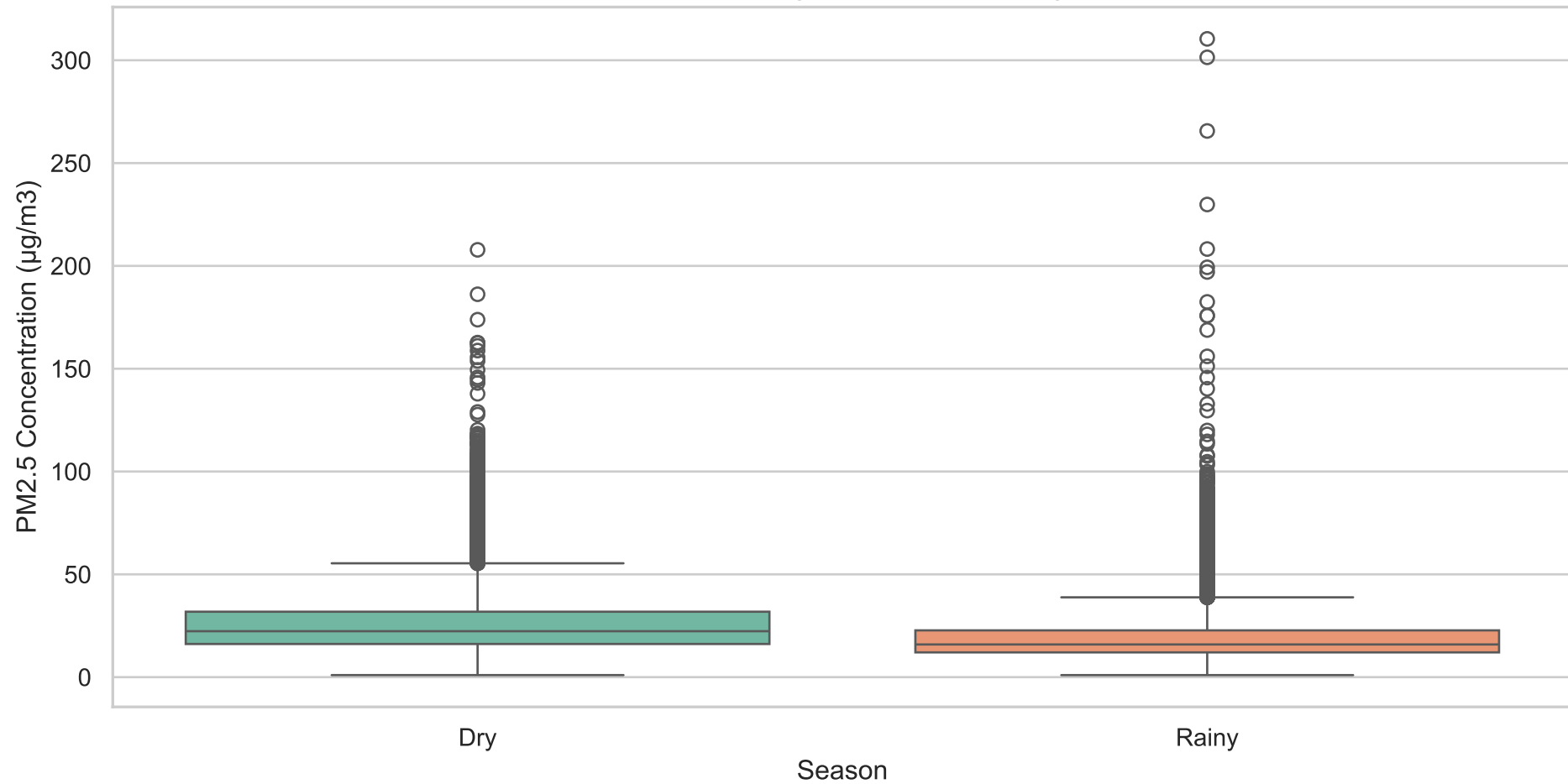


Explanation: PM2.5 is right-skewed (Mean > Median). Most days are 'Good' or 'Moderate', but extreme values drive the mean up.

Air Quality Index (AQI) Categories

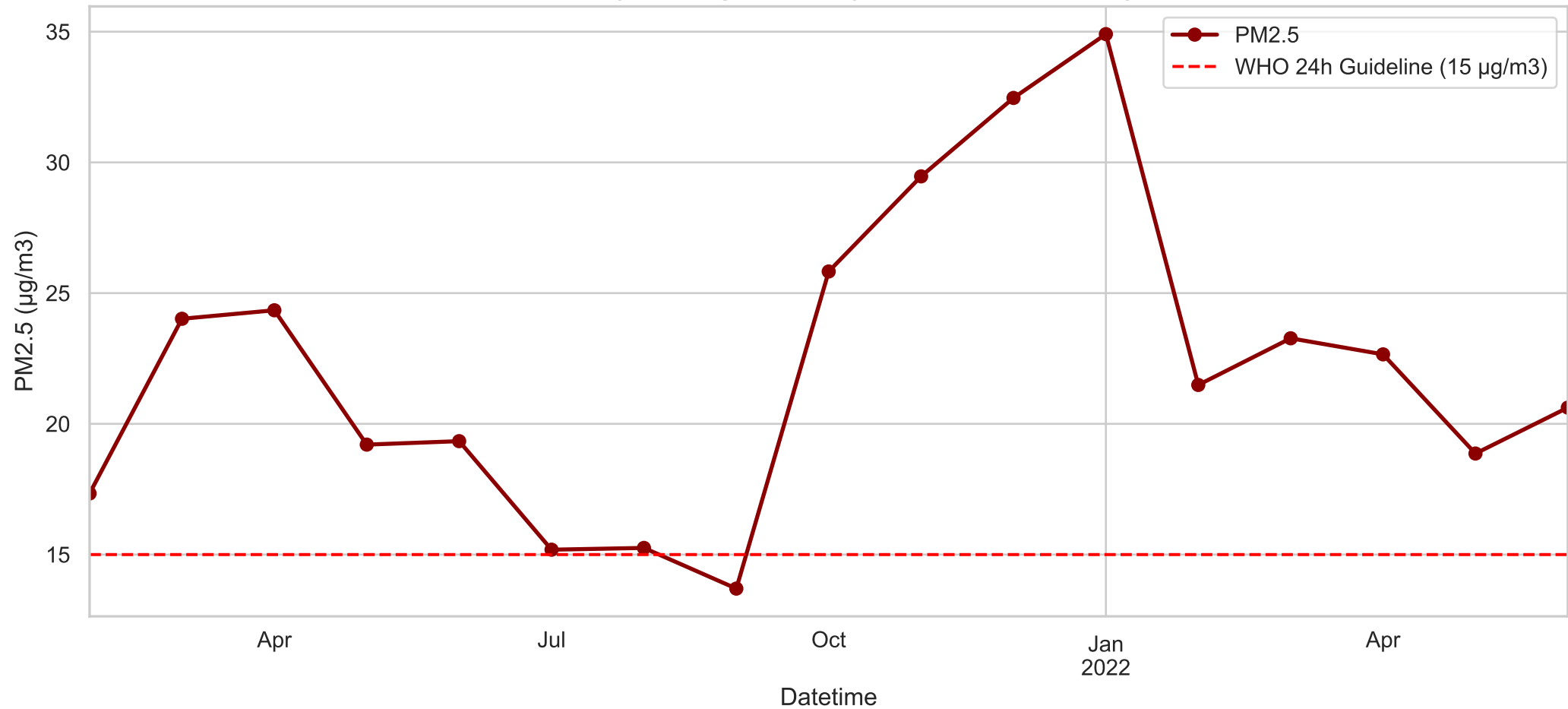


Seasonal Impact: Dry Season vs Rainy Season



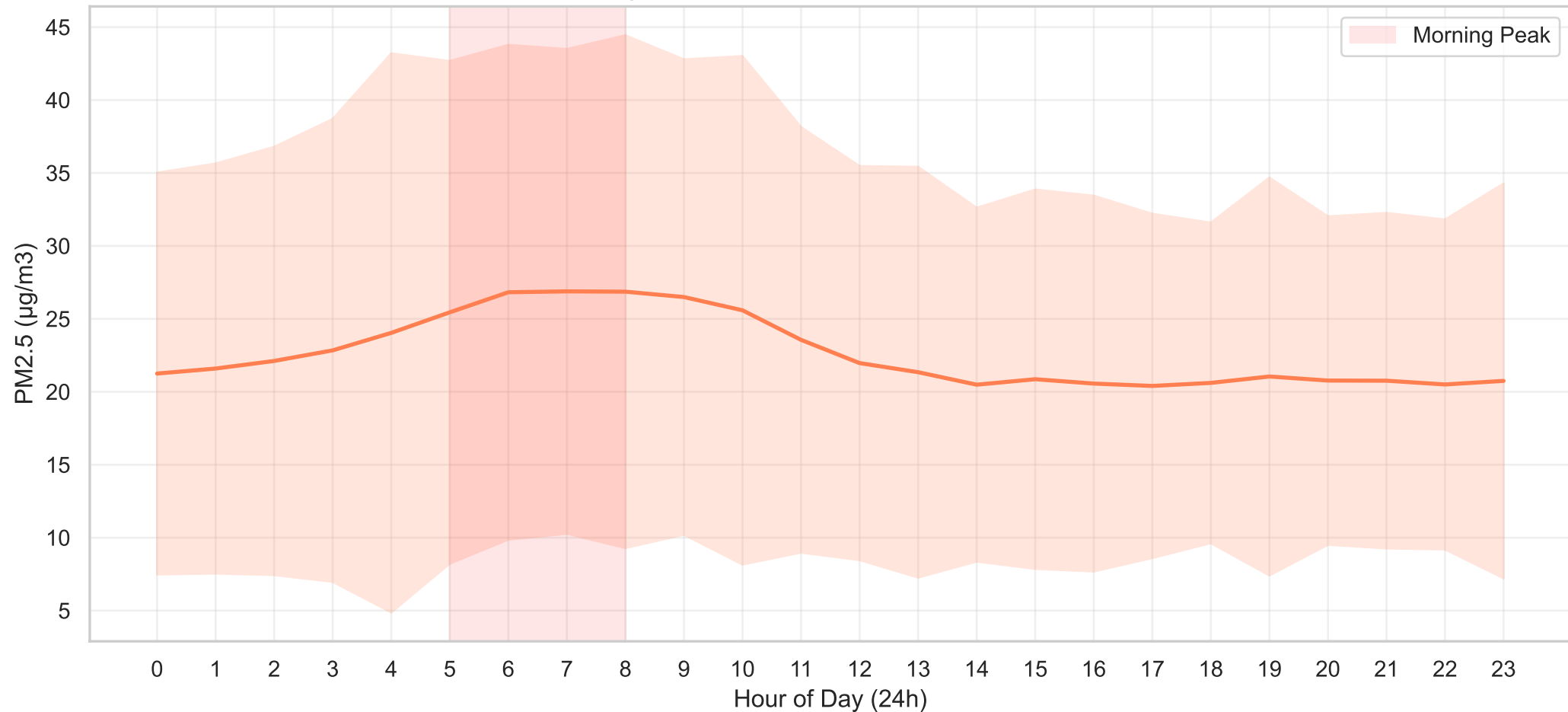
Insight: Dry season (Nov-Jan) pollution levels are roughly DOUBLE those of the rainy season (Jul-Sep). Rain acts as a natural scrubber, removing pollutants from the atmosphere.

Monthly Average PM2.5 (Feb 2021 - Jun 2022)



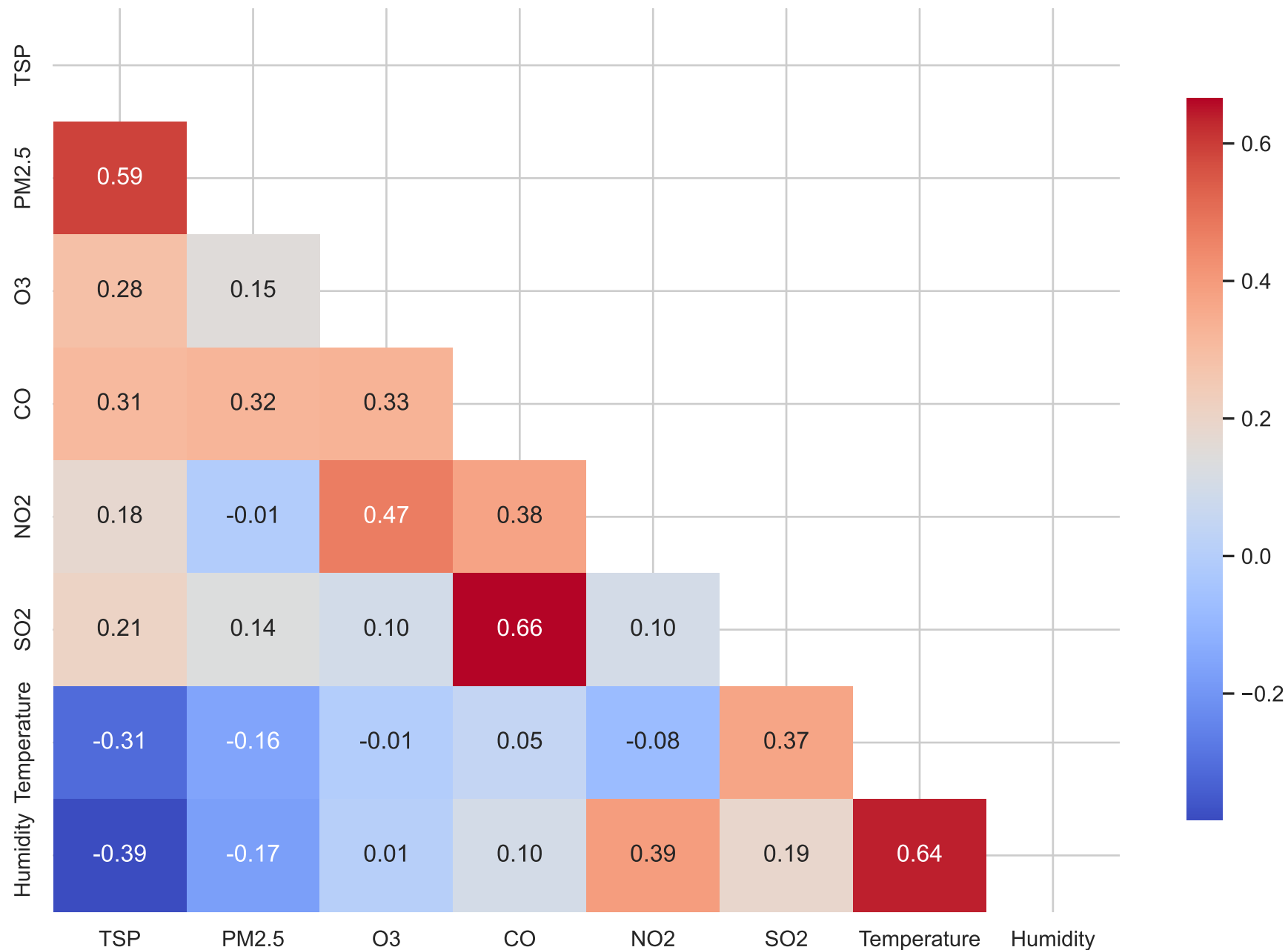
Observation: Pollution levels frequently exceed WHO guidelines, especially during the transition into the dry season.

Hourly Variation: The "Rush Hour" Effect



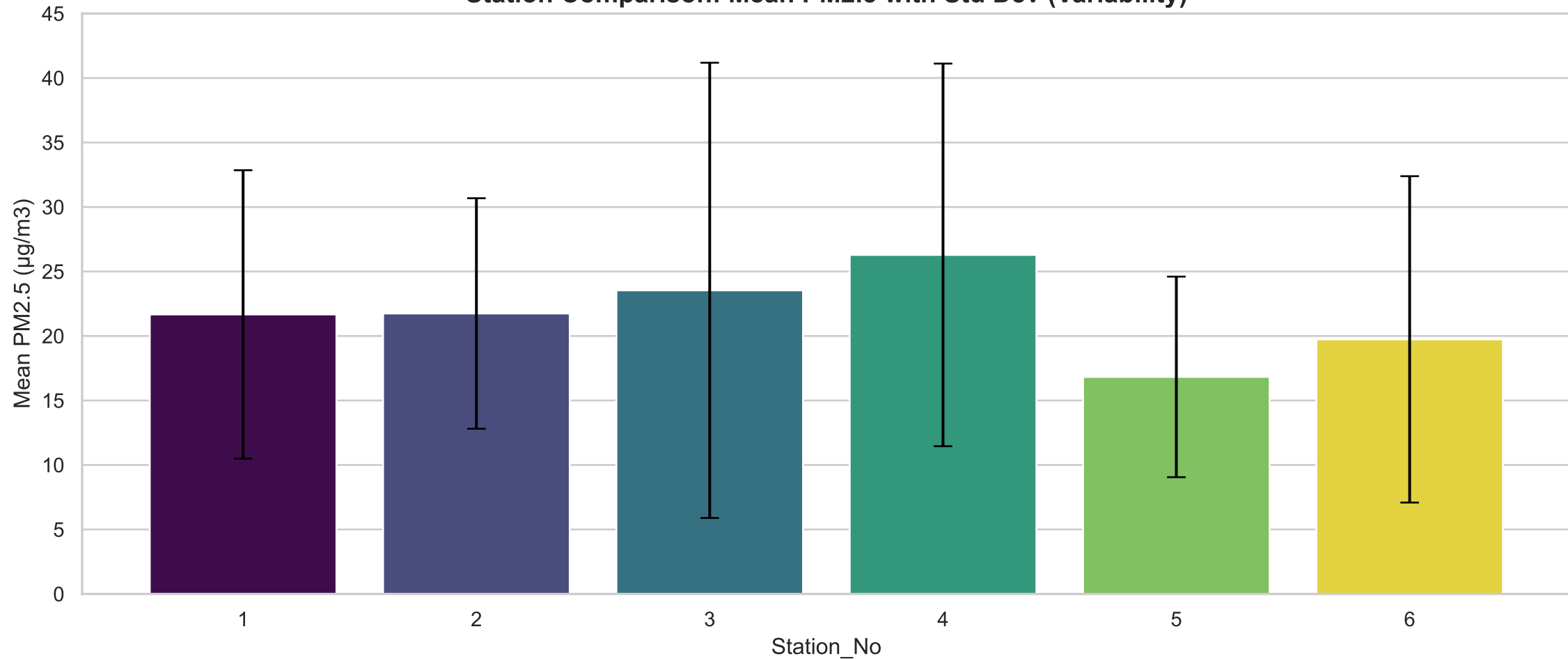
Insight: PM2.5 is highest between 5-8 AM. This is caused by a combination of morning traffic and a 'stable boundary layer' that traps pollutants near the ground.

Pollutant & Weather Correlation Matrix



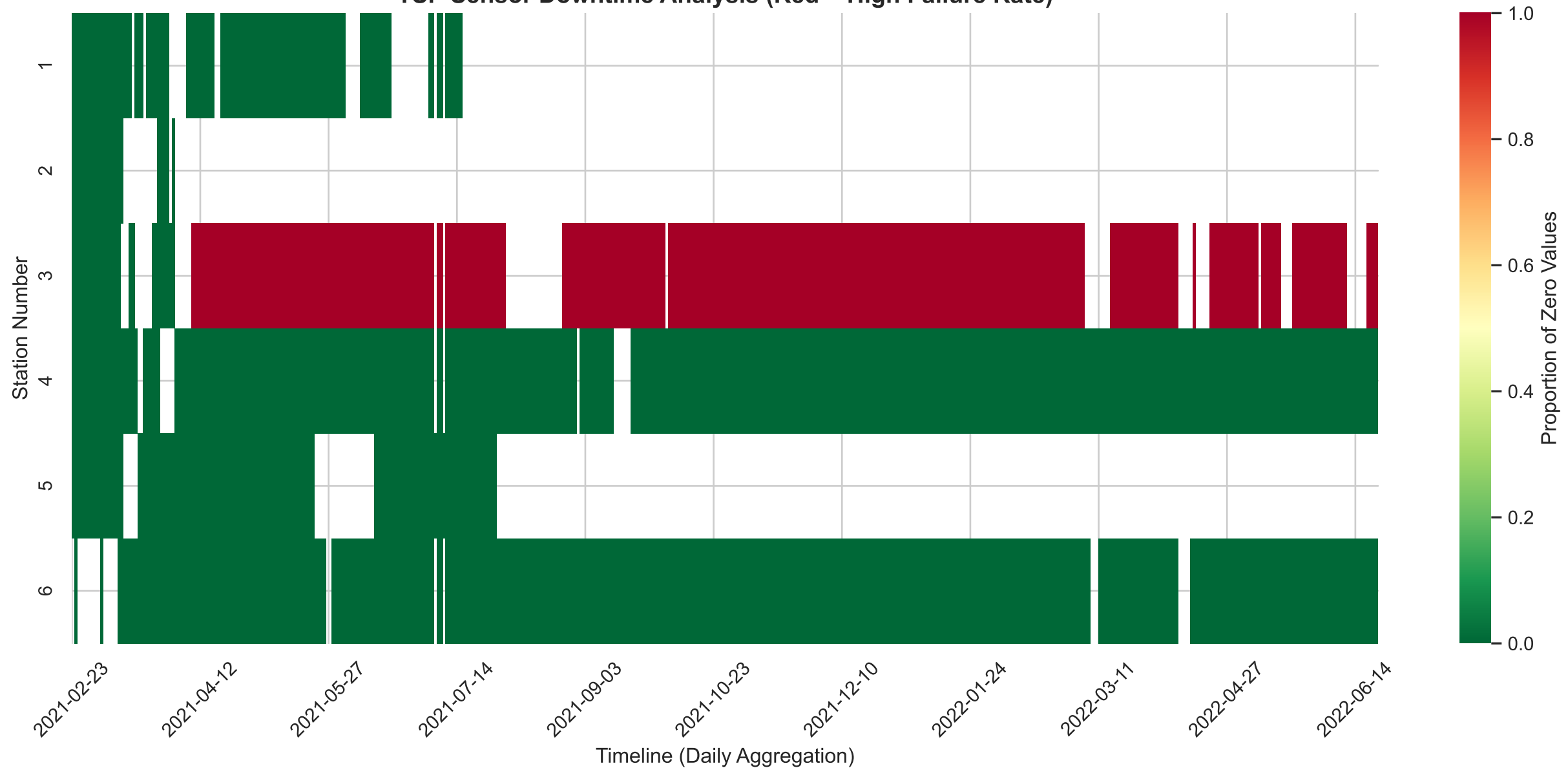
Analysis: Strong correlation (0.66) between CO and SO2 suggests they share common emission sources, likely vehicle exhaust or industrial combustion.

Station Comparison: Mean PM2.5 with Std Dev (Variability)



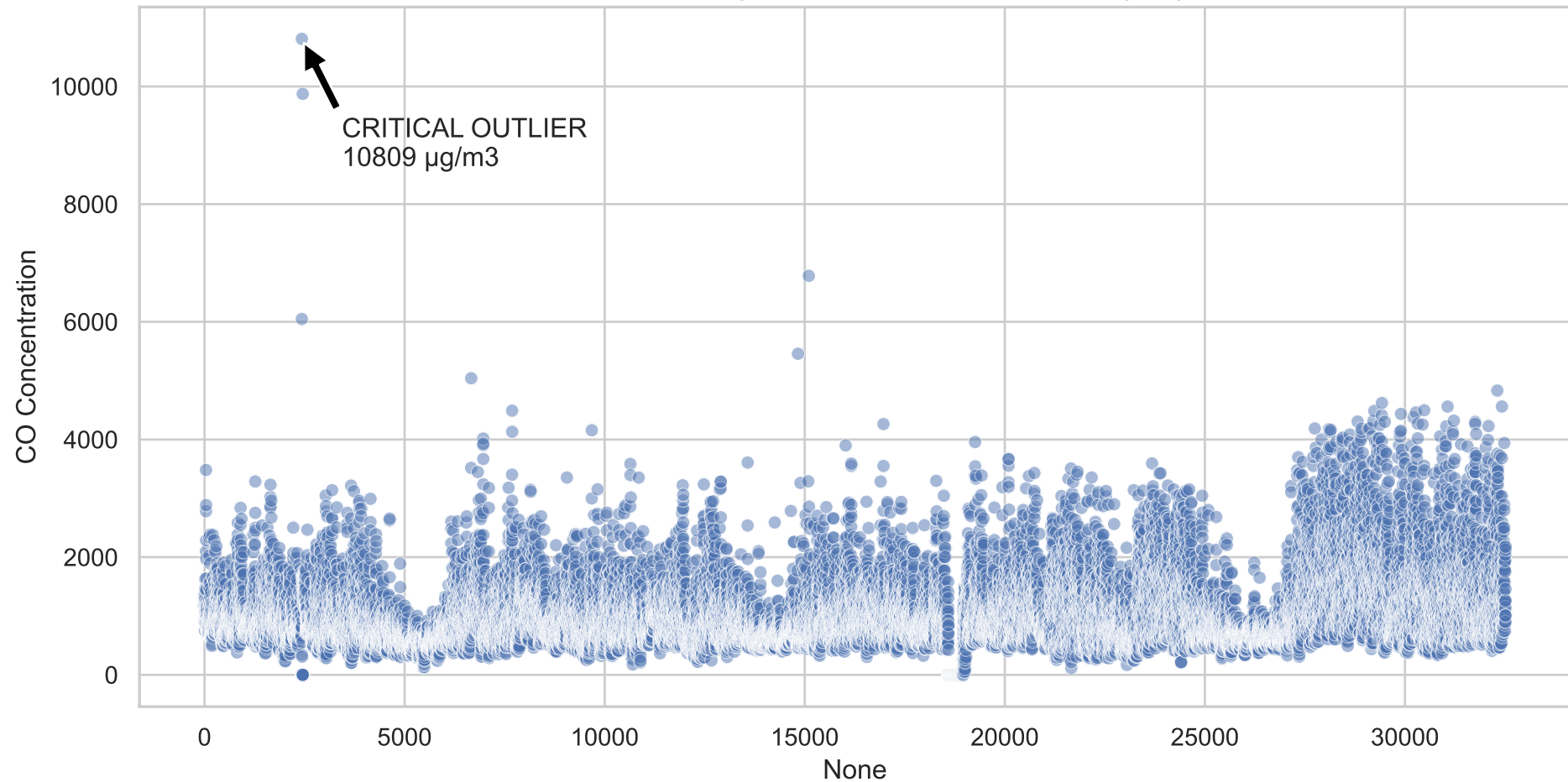
Warning: Station 4 is most polluted (Mean 26.3). Station 3 shows extreme variability (Std=17.65), suggesting unstable conditions or localized events.

TSP Sensor Downtime Analysis (Red = High Failure Rate)



Interpretation: Solid Red blocks indicate extended periods of sensor failure (Station 3).
Green areas represent healthy, non-zero data collection.

Outlier Investigation: Carbon Monoxide (CO)



Alert: The extreme CO value (10,809) at Station 2 is highly suspicious. It is 27 times the average and likely represents a sensor malfunction.

IMPLICATIONS FOR MODELING (Next Steps)

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1. DATA TRANSFORMATION:

- Use Log-Transformation for PM2.5 to handle right-skewness.
- Normalize meteorological variables (Temp/Humidity) for model stability.

2. FEATURE ENGINEERING:

- Create "Is_Weekend" and "Rush_Hour" flags based on time analysis.
- Add Lag-features (PM2.5 at $t-1$, $t-2$) to capture temporal dependency.

3. OUTLIER HANDLING:

- Remove the CO record (10,809) from Station 2.
- Impute or exclude TSP records where sensor was offline (zeros).

4. STATION HANDLING:

- Focus modeling on Stations 3, 4, 5, 6 for long-term consistency.
- Consider One-Hot Encoding for Station_No if training a global model.